

Private 5G for real-time energy monitoring and optimization in French manufacturing plants to support reindustrialization and sustainability goals.

Manufacturing × Energy optimisation × Private 5G / LTE · OS163 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
69 is the world moving	76 can we play, can we win	HIGH 0 named in the evidence	€152m partial evidence · per year	NOW budgeted procurement within 1...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

Private 5G is the invisible foundation for French reindustrialization, enabling real-time energy monitoring and optimization in manufacturing plants. This opportunity targets French manufacturers seeking to meet sustainability goals while improving operational efficiency. The time is now, as procurement is budgeted within 12 months.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€214m €27.2m – €1.4bn	€152m €19.2m – €991m	€7.3m €924k – €47.6m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (FR)	20,243 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Private 5G / LTE	20.1 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	3,155 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Private 5G / LTE: No private-network series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

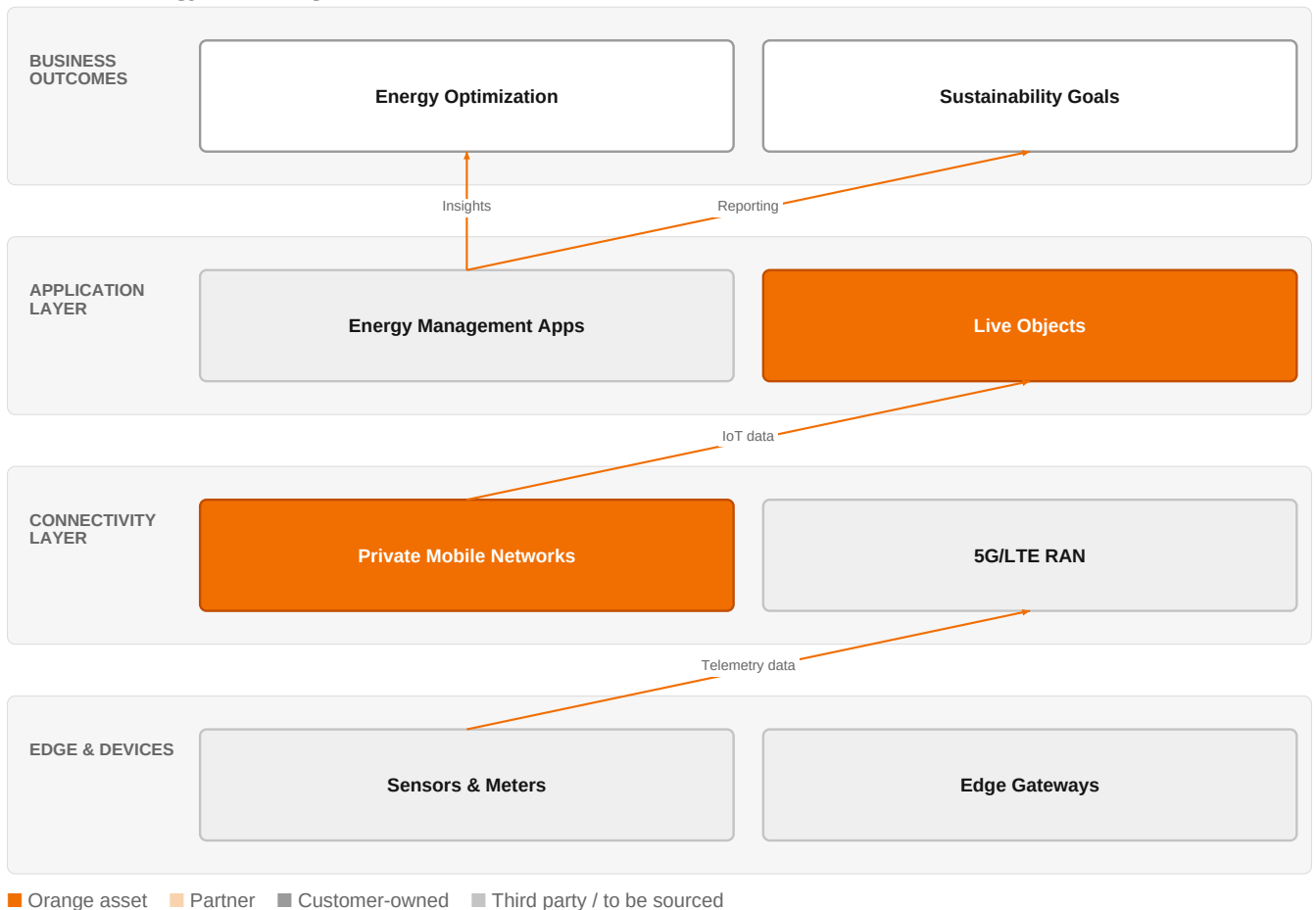
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Private 5G Energy Monitoring Solution



This diagram shows how private 5G connects sensors to energy management applications, enabling real-time optimization and sustainability reporting.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Private 5G/LTE networks are increasingly recognized as essential for Industry 4.0, offering deterministic reliability and ultra-reliable low-latency communication (URLLC) that Wi-Fi cannot match. The integration of private 5G with mobile edge computing (MEC) and network slicing enables tailored, latency-sensitive services for smart manufacturing. French reindustrialization efforts are positioning private 5G as a foundational technology, with public tenders for energy management and renovation projects signaling a shift toward energy efficiency and sustainability.

Sources: SIG-4231EEBFD758 openalex.org 2026-01-27; SIG-873E0DC9CA64 International Journal 2025-11-07; SIG-AF0B9D624A57 arxiv.org 2026-06-30; SIG-149B67B9DA26 InformatiqueNews.fr 2026-04-29; SIG-A112E6B4A541 ted.europa.eu 2026-07-08

Who buys, and why now

The plant operations director or site manager feels the pain of rising energy costs and sustainability targets. The IT director signs for the network infrastructure, while the CFO approves the budget. The trigger is often a regulatory deadline or a corporate sustainability mandate that requires real-time energy visibility and optimization. They need to reduce energy waste, improve equipment efficiency, and report on carbon footprint, but current wireless networks lack the reliability and determinism required for real-time monitoring and control.

Why it is hot now — every claim bound to a dated source

- Private 5G is positioned as the invisible foundation for French reindustrialization. [SIG-149B67B9DA26]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a private mobile network (using the 'Private mobile networks' offer) tailored to the plant's energy monitoring needs. The engagement begins with a site survey and network design, leveraging Orange's 'IoT experts' and 'Digital experts' to integrate sensors and energy management systems. The network would be deployed with 'Live Objects' for IoT data management, enabling real-time energy analytics. Orange's 'Defense and security experts' ensure the network meets security standards (ISO 27001, TISAX). The solution is delivered as a managed service, with ongoing optimization and support.

Why Orange can win

Orange is a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services, demonstrating market recognition. Orange has proven experience with manufacturing clients such as Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass, and holds relevant certifications (ISO 27001, TISAX). The combination of private mobile networks, IoT expertise, and security capabilities provides a comprehensive solution that competitors may lack.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer addresses the use case but not this technology	unconfirmed
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
Samsung	partner	L2	partner provides the required technology	unconfirmed
Huawei	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader)	analyst position	SUP	analyst recognition covers this technology or use case	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 8.0; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
BT Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Private 5G / LTE; active in Manufacturing; competes in Connectivity Solutions, OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	sells Private 5G / LTE; competes in Connectivity Solutions, OX: Smart Industries	structural	—

Proximus	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Swisscom	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telia Company	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What are your current energy monitoring and optimization challenges in the plant?
- 2 Do you have any existing IoT or sensor infrastructure for energy management?
- 3 What are your sustainability targets and timelines?
- 4 Have you considered private 5G/LTE for your plant connectivity?
- 5 What is your budget and procurement timeline for this initiative?
- 6 Who is the key decision-maker for this project?

Objections you will meet

They will say	You can answer
We already have Wi-Fi; why do we need private 5G?	Wi-Fi can be unreliable and lacks the determinism needed for real-time energy monitoring. Private 5G offers ultra-reliable low-latency communication, ensuring consistent performance.
Private 5G seems expensive.	While the initial investment may be higher, the long-term energy savings and operational efficiency gains can offset the cost. Orange offers managed services that can reduce total cost of ownership.
We are concerned about security.	Orange's solution is built with security in mind, holding ISO 27001 and TISAX certifications. Our defense and security experts ensure your network is protected.

Risks and unknowns

The main risk is that the customer may not see the immediate ROI of private 5G versus Wi-Fi or public 5G. There is also uncertainty about the availability of spectrum and regulatory approvals. Additionally, the integration with existing energy management systems may be complex, and the customer may have legacy equipment that is not compatible. The sales team must be prepared to demonstrate clear business value and address these concerns.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a French manufacturer, differentiating Orange's private 5G and Live Objects offering by citing the Gartner Leader position, ISO 27001 certification, and the Saint-Gobain Glass reference, while noting TISAX for automotive prospects.
Sales	In a meeting with a French manufacturing plant's operations director, say: 'Orange's private 5G network, combined with our Live Objects IoT platform, can give you real-time energy visibility to support your sustainability targets, as we've done for Saint-Gobain Glass.'
Strategist / Innovator	Commission a deep-dive brief on how Orange's Private mobile networks and Live Objects, backed by the Gartner Magic Quadrant 2026 Leader position, can be packaged for real-time energy monitoring in French manufacturing, leveraging IoT and Digital experts to validate the value proposition.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FC4FCB6FF0CC	2026-08-05	ted.europa.eu	1	Italy – Energy-management services – Gara europea per l'affidamento in concessione, tramite fin...
SIG-A112E6B4A541	2026-07-08	ted.europa.eu	1	France – Energy and related services – 26552 - Accord-cadre d'accompagnateur Thermique pour la ...
SIG-AC86A5D91D50	2026-06-11	ted.europa.eu	1	France – Pollution tracking and monitoring and rehabilitation – SPCH a Harbonnieres (80) - Etud...
SIG-072593234114	2026-06-01	cordis.europa.eu	1	Pilot for Resilient Industrial Manufacturing Environments with 6G Technologies
SIG-4231EEBFD758	2026-01-27	openalex.org	1	Software-based 5G industrial wireless private networks :
SIG-9798339F3F2D	2026-01-01	IEEE Access	1	Multi-Connectivity for Industrial Automation: Achieving Deterministic Reliability Through Priva...
SIG-873E0DC9CA64	2025-11-07	International Journal of ...	1	Architecting and Analyzing Private 5G/LTE Networks for Ultra-Reliable and Deterministic Industr...
SIG-6FBA261C5571	2025-08-08	International Journal on ...	1	Security Challenges and Solutions in Private Wireless (LTE/5G) Network Implementations
SIG-149B67B9DA26	2026-04-29	InformatiqueNews.fr	2	5G privee, le socle invisible de la reindustrialisation francaise - InformatiqueNews.fr
SIG-77F434506B29	2026-07-29	framerusercontent.com	3	Industry Brief: Private 5G for Manufacturing and Industrial Sites [pdf]
SIG-1B48F0661755	2026-07-15	framerusercontent.com	3	Industry Brief: Private 5G for Events and Venues [pdf]
SIG-A57E9B478F46	2026-07-01	arxiv.org	3	5G Configured Grant Scheduling for 5G-TSN Integration for the Support of Industry 4.0
SIG-AF0B9D624A57	2026-06-30	arxiv.org	3	Latency-Sensitive 5G RAN Slicing for Deterministic Aperiodic Traffic in Smart Manufacturing

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:35+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.